

Spring Boot

Basic Operations on Spring Boot :-

1. JSP Include Directive: `<%@ include %>`

What it does -

- `<%@ include %>` is a static include in JSP.
- It inserts the content of another JSP file at compile time (before the page is translated into a servlet).

Example -

```
<%@ include file="header.jsp" %>
```

This means -

- Entire content of header.jsp is copied into the current JSP.
- It behaves just like you wrote all header.jsp code inside your page.

Use cases -

- Common layout parts (header, footer, navbar)
- Reusable components across pages
- Better maintainability

2. Context Path: `${pageContext.request.contextPath}`

What is context path?

- The context path is the base path of your web application.

Examples –

Application URL	Context Path
http://localhost:8080/about	/about
http://localhost:8080	(empty, root)

Why use it?

- Because your project might not always run at root (/).
- When you deploy Spring Boot:
 1. On Tomcat as WAR
 2. Behind a proxy

3. Under a sub-folder

the base URL changes.

Example -

```

```

This ensures:

/MyApp/resources/images/logo.png

Static Resources With Spring :-

- Spring Boot serves static content automatically.
- Static content includes CSS, JavaScript, and images.
- Files are served from specific project locations.
- The browser loads these files when rendering JSP or HTML.

1. Default Static Resource Locations -

Spring Boot looks for static files in these folders by default:

```
src/main/resources/static/  
src/main/resources/public/  
src/main/resources/resources/  
src/main/resources/META-INF/resources/
```

Most commonly used: src/main/resources/static/

2. Folder Structure Example -

```
src/main/resources  
├── static  
│   ├── css  
│   │   └── mystyle.css  
│   ├── images  
│   │   └── marek-piwnicki-TKXGgqnfxEO-unsplash.jpg  
│   └── js  
│       └── myscript.js
```

- CSS files → css/
- Images → images/
- JavaScript files → js

3. URL Mapping –

- With Spring Boot, static resources are automatically available under / path.

Example:

File Location	URL in Browser / JSP
static/css/mystyle.css	/css/mystyle.css
static/js/script.js	/js/script.js
static/images/logo.png	/images/logo.png

4. Customizing Static Resource Locations -

You can change the default location or add a URL prefix in application.properties:

```
spring.web.resources.static-locations=classpath:/static/  
spring.mvc.static-path-pattern=/resources/**
```

- **spring.web.resources.static-locations** → tells Spring Boot where the static files are located.
- **spring.mvc.static-path-pattern** → the URL pattern used to access the static files.

Implementation in SpringBoot :-

SpringBootSampleProject1Application.java -

```
package com.example.demo;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class SpringBootSampleProject1Application {

    public static void main(String[] args) {
        SpringApplication.run(SpringBootSampleProject1Application.class, args);
    }

}
```

MyController.java -

```
package com.example.demo;

import org.springframework.stereotype.Controller;
import org.springframework.web.bind.annotation.GetMapping;

@Controller
public class MyController {

    @GetMapping("/")
    public String home()
    {
        return "home";    //home.jsp
    }

    @GetMapping("/about")
    public String about()
    {
        return "about";    //about.jsp
    }

}
```

home.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
<style>
    h3{
        color:red;
    }
</style>

    <link rel="stylesheet" href="${pageContext.request.contextPath}/resources/css/mystyle.css" type="text/css">
    <script src="${pageContext.request.contextPath}/resources/js/myscript.js"></script>

</head>
<body>
    <%@ include file="header.jsp" %>

    <h2 style="color:green">Welcome To Spring Boot</h2>
    <h3>Good Evening</h3>
    <p>Have a nice Day!</p>

    <button type="button" onClick="test()">Click Here</button>
    <br><br>
    

</body>
</html>
```

about.jsp –

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>

</body>
</html>
```

header.jsp –

```
<nav>
    <a href="/">Home</a>
    <a href="/about">About</a>
</nav>
```

mystyle.css –

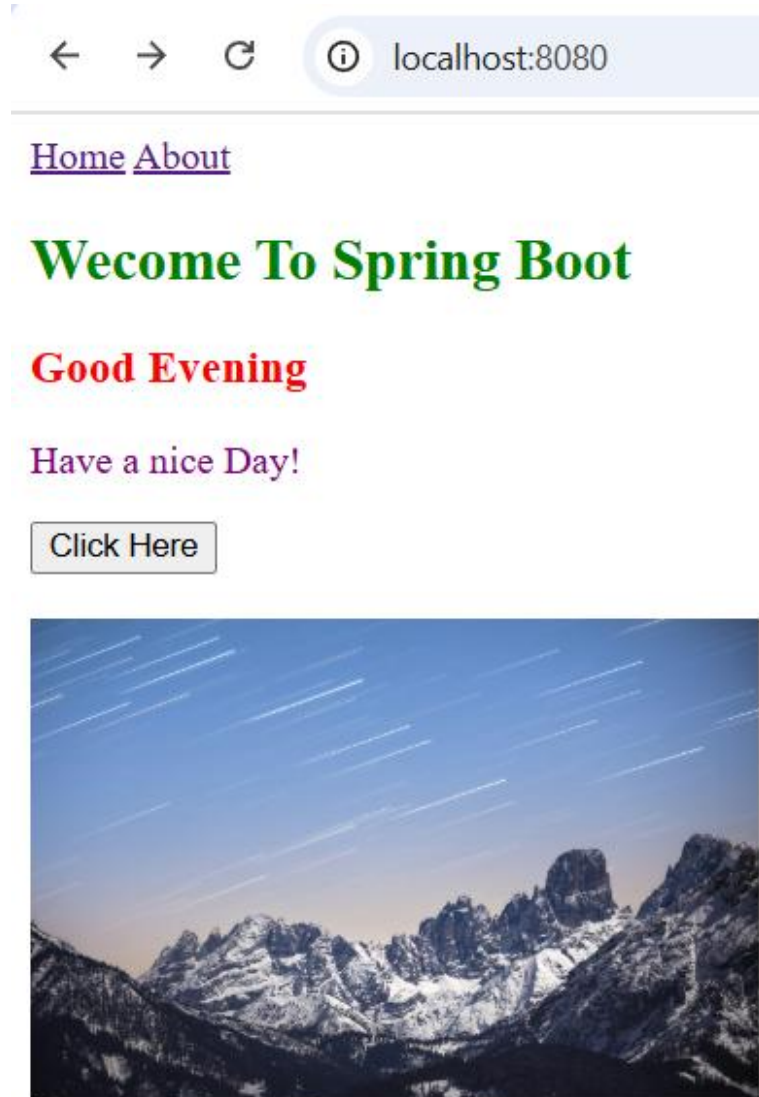
```
p{
    color:purple;
}
```

myscript.js –

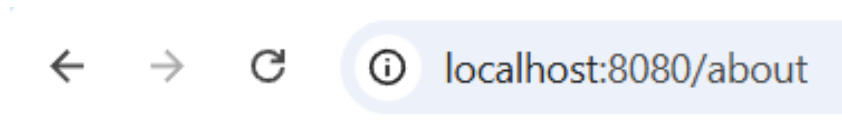
```
function test()  
{  
    alert("I am test function");  
}
```

Output –

Home Page -



About Page –



Note : The Blank Page about.jsp they do not return any inside the body section.

After Click on Button then –

